Final Project — Stretch Robot Proof of Concept

Robotics and Cyber-Physical Systems
School of Sciences and Engineering
Tecnológico de Monterrey

Overview

For the final project, your team will identify a practical use case for the Hello Robot Stretch and implement a proof-of-concept demonstration that shows the idea is feasible given the robot's hardware.

The goal is not a finished, production-ready system. It is a convincing demonstration: something that works well enough to make a clear case that the application could be built out further. Think of it as pitching an idea to a potential collaborator or investor—you want to leave them believing the concept is real and achievable.

The `stretch_mujoco_digital_twin` simulator is your primary development platform. Scripts written against `stretch_toolkit` are designed to port to the physical Stretch robot with minimal changes.

Choosing a Use Case

There are no restrictions on the application domain. Any use case that makes genuine use of the Stretch robot's capabilities is acceptable. Some directions to consider:

- Assistive robotics (helping people with limited mobility perform everyday tasks)
- Object manipulation and sorting
- Inspection or monitoring tasks
- Human–robot interaction scenarios
- Navigation and fetch tasks

Your use case should be specific enough that you can scope a meaningful proof of concept within the project timeline. Broad ideas ("a general-purpose home robot") are harder to demonstrate convincingly than focused ones ("a robot that retrieves a specific object from a shelf on request").

Deliverables

Every team must submit the following:

1. **Source code** — A working script (or set of scripts) that runs in the simulator. Code should be reasonably clean and include brief comments explaining non-obvious sections. Submit via the course platform or as a link to a repository.
2. **Written report** (2–3 pages) — A concise document covering:

- The use case and why the Stretch robot is a suitable platform for it
- A description of how the system works
- Known limitations and what would be needed to take it further

3. **Video** — A recording that shows your demo running. This can be a raw screen capture or a narrated/edited video—whatever best conveys what the system does. The video should be self-explanatory to someone who has not seen your presentation.

File naming: `Team_[TeamName]_FinalProject.zip` containing all of the above.

Presentation

Each team will present to the class. The presentation should include:

- **Slides** — Cover the use case, your approach, and the key design decisions.
- **Demo** — Show your system running live in the simulator. If a live demo is not practical on the day, a pre-recorded video of the simulation is acceptable as a fallback.
- **Q&A** — Be prepared to answer questions about your implementation and design choices.

Presentations should be roughly 10–15 minutes including Q&A.

Bonus Opportunity

Teams that successfully run their proof-of-concept on the **physical Stretch robot** are exempt from the class presentation. You still need to submit all three written deliverables (code, report, video).
To arrange a session, use the scheduling tool provided by the instructor to book a time slot. Not every team member needs to attend—one or two representatives are sufficient. Multiple teams are welcome to coordinate and share a single session, taking turns with the robot.

Grading

The project is graded across four equally-weighted areas:

Area	Weight	What we are looking for
Idea & Feasibility	25%	Does the use case make sense for this robot? Is the scope appropriate—ambitious enough to be interesting, bounded enough to demonstrate?
Implementation	25%	Does the demo work? Does the code make reasonable use of the robot's capabilities? Is there some autonomous behavior, or is it purely teleoperated?
Presentation	25%	Is the pitch clear and convincing? Does the demo (live or recorded) support the argument? Can the team answer questions about their system?
Video	25%	Does the video clearly show what the system does? Would someone watching it understand the idea without additional explanation?

Note

These criteria are intentionally broad. There is no single correct way to build a proof of concept, and grading will take into account the difficulty and ambition of what you attempted relative to what you achieved.